Vélib' project: raw data quality and cleaning (one page)

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Detail and evidence: `data_quality_and_cleaning.pdf`.

Data. The GBFS `station_status` feed, captured every 5 minutes by two Azure VMs since 20.09 (45 days). Figures are from 21.09, the complete day shipped in `raw_sample/` (437,184 rows, 1,518 stations), unless marked *full archives*.

Quality of the raw data

Level	What we found
Collection	288 of 288 captures every day on both VMs over the 43 full days (*full archives*); each VM holds every slot, so merging them adds none. No failed request and no gap over 10 minutes up to the 29.09 snapshot.
File content	0.8% of captures are partial (96 on VM1, 98 on VM2, *full archives*): bike and dock counts present, `stationCode` and mechanical / electric split missing. 96 fall in the same slot on both VMs, so the cause is the server, not us. No truncated capture in 295 checked files.
Reliability of our own capture	On 307 slots caught by both VMs, the median share of stations with identical counts is 100% (minimum 75.8%): a capture is close to instantaneous. No exact duplicate on one machine; some will appear when the two archives are merged.
Meaning	`last_reported` is not the time of the counts: it renews about once an hour, and 90.5% of count changes happen while it is unchanged.
Stations	Concorde and Invalides (about 410 docks, open one day) appear for the 26.09 event; 4 stations with capacity 0; 0.85% of rows with bikes + docks above capacity; 23 silent and 3 frozen stations. 1.6% of rows are out of service, 26 stations all day: a real failure for users, so kept.

Main cleaning choices, and why

We **flag, we do not alter**: the feed is our only source, so nothing can correct a count. Rows are dropped only when they carry no information; every other problem is a boolean column, and `in_panel` combines them so each exclusion stays visible and reversible.

1. **Drop** captures with fewer than 1,400 stations (network: about 1,519) and repeated `(station_id, fetched_at)` rows. **Keep** partial captures, since availability only needs the counts.
2. **Match each day with that day's station list**, because the network changes during the period.
3. **Exclude temporary and event stations** (100+ docks) from the panel: they would count as failures or as false alternatives.
4. **Keep every capture, no deduplication on `last_reported`.** At one capture per hour, or with our old plan, only about half of the 5-minute episodes remain visible (48.7% and 52.4%) and the share of time without a bike drops from 6.67% to 5.82%.
5. **Flag** silent (`last_reported` older than 180 min) and frozen stations (identical counts for 12 h or more); capacity 0 becomes unknown, capacity conflicts are flagged, not corrected.

Result on 21.09: the panel keeps 435,910 of 437,184 rows (99.7%) and 1,515 stations.

What the cleaned day already shows (21.09)

Measure	Value
Share of time a station cannot rent / cannot take a return	6.7% / 4.7%
Failure episodes (at least 2 consecutive captures)	2,602 no-bike and 1,639 no-dock, plus 28 per side lasting the whole day, almost all at stations out of service
Median observed duration	25 min (no bike), 20 min (no dock); 56% of no-bike episodes last under 30 min

Durations are what we saw: a real episode can be up to 5 minutes longer at each end.

Storage

On the cleaned day, Parquet partitioned by date and read with DuckDB takes 1.8 MB for 29 columns, against 46.8 MB for 8 columns in SQLite, and scans faster (10 ms vs 63 ms for the share of time without a bike). SQLite is faster for one-station lookups (0.4 ms vs 7 ms). Our queries are mostly scans, so we keep Parquet, and will measure again on the whole period.

Limits and what is still open

- Eight weeks of autumn: no winter, no summer. Demonstrations in Paris after the 29.09 snapshot closed or damaged some stations; for the final project we will mark those days and report the results with and without them.
- The counts are renewed at least every 5 minutes, but the feed does not say how long after an event they change, so a short delay may remain.
- Not yet done: the cleaning run on the whole period (our archive is too large to ship), and the commune boundaries (IGN) and population (INSEE), joined in the analysis step as context only.